Amrita School of Engineering, Bengluru-35
23MAT112
Mathematics for Intelligent Systems – 2
Lab Practice Sheet-5
(Probability – pdf, cdf, icdf)

- Evaluation of pmf (f(x)=P(X=x)) and cdf (F(x)=P(X≤x)) for standard discrete distributions:

  Binomial:

  >> pdf('bino', x, n,p)

  %finds f(x) =P(X=x) in binomial distribution with n trials and probability of success, p.

  >> cdf('bino', x, n,p)

  %finds F(x) =P(X≤x) in binomial distribution with n trials and probability of success, p.

  Poisson:

  >> pdf('pois', x, lambda) or

  %finds f(x) =P(X=x) in Poisson distribution with parameter lambda.

  >> cdf('pois', x, lambda)

  %finds F(x) =P(X≤x) in Poisson distribution with parameter lambda.

- Evaluation of cdf (F(x)=P(X≤x)) for standard continuous distributions:

  Continuous Uniform:

  >> cdf('unif', x, a,b)

  or

  >>unifcdf(x, a,b)

  %finds F(x) =P(X≤x) for uniform rv X with pdf, f(x)=1/(b-a), a<x<b

  Exponential

  >> cdf('exp', x, mean)

  Or

  >> expcdf(x, mean)

  %finds F(x) =P(X≤x) for Poisson rv X with given mean.

  Normal:

  >> cdf('norm', x, μ, σ)

  Or

  >> normcdf(x, μ, σ)

  %finds F(x) =P(X≤x) for normal rv X with mean μ and standard deviation, σ.

Examples:

```
%Binomial pdf
>> pdf('bino',2,4,0.1) % f(2) for n=4, p=0.1
ans =
   0.0486
```

```
%Binomial cdf
>> cdf('bino',2,4,0.1) % F(2) for n=4, p=0.1
ans =
   0.9963
```

```
%Poisson pdf
>> pdf('pois',5,5) % f(5) for λ=5
ans =
   0.1755
```

```
%Poisson cdf
>> cdf('pois',2,5) % F(2) for λ=5
ans =
   0.1247
```

```
%Exponential cdf
>> cdf('exp',30000,25000) % Evaluates F(30000) for exponential distrn with mean 25000
ans =
   0.6988
```

```
% Normal cdf
>> cdf('norm',290,300,50) % Evaluates F(290) for normal distrn with mean 300 and s.d. 50
ans =
   0.4207
```

- **Use of 'icdf' to find 'a' if cumulative function at a, F(a) is given**

  >> a = icdf('D', F(a), P1, P2)

  is used to find the value of 'a' for the distribution 'D', given the cdf at a is the probability value, F(a) and P1 and P2 are the parameters of the distribution D.

  Examples:

  1. Given P(X<a)=0.85, where X is normally distributed with mean 9 and standard deviation 2. Find 'a'.

     >>a = icdf('norm',0.85,9,2)

     a = 11.0729

  2. Given P(X>b)=0.35, where X is an exponential random variable mean 5. Find b.

     P(X>b)=1-F(b)=0.35

     So F(b)=0.65

     >>b = icdf('exp',0.65,5)

     b = 5.2491

  3. Given P(-c < X < c) = 0.8732 where X is a standard normal random variable. Find c.

     P(-c < X < c) = 0.8732 → 2*P(0<X<c) = 0.8732 (since std normal is symmetric about the origin)

     →2 * (F(c) - F(0)) = 0.8732

     → 2 * (F(c) – 0.5) = 0.8732

     → 2 * F(c) = 1.8732

     → F(c) = 0.9366

     >>c = icdf('exp',0.9366,0,1)

     c = 1.5268

- **Plot of pdf and cdf of probability distributions:**

```
%Plot of pdf of exponential distribution with lambda=0.5 (mean=2)
clf;clc;
x=0:0.1:10;
mean=2;
f=exppdf(x,mean);
plot(x,f)
hold on
F=expcdf(x,mean);
plot(x,F,'r')
```

```
%Plot of pdf of uniform distribution with a=2, b=7
clf;clc;
x=0:0.1:15;
a=2;
b=10;
f=unifpdf(x,a,b);
plot(x,f)
F=unifcdf(x,a,b);
plot(x,F)
```

```
%Plot of pdf of normal distribution with mu=3, sigma=1
clf;clc;
x=0:0.1:10;
Mu=3;
Sigma=1;
f=normpdf(x,Mu,Sigma);
plot(x,f)
F=normcdf(x,Mu,Sigma)
plot(x,F)
```

```
%Plot of pdf of standard normal distribution in [-5,5]
clf;clc;
x=-5:0.1:5;
Mu=0;
Sigma=1;
f=normpdf(x,Mu,Sigma);
plot(x,f)
F=normcdf(x,Mu,Sigma)
plot(x,F)
```

Practice Problems:

1. If X is a Binomial random variable with n=10, probability of success, p=0.3, find P(X=5), P(X=10), P(X≤2), P(X>8).
2. The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline.
    a. What is the probability that for exactly three calls the lines are occupied?
    b. What is the probability that for at least one call the lines are not occupied?
    c. What is the expected number of calls in which the lines are all occupied?
3. Human error is given as a reason for 75% of all accidents in a plant. Use the formula for the binomial distribution to find the probability that human error will be given as the reason for 2 of the next 4 accidents.
4. An agricultural cooperative claims that 90% of the watermelons shipped out are ripe and ready to eat. Find the probabilities that among 18 watermelons shipped out,
    a. all 18 are ripe and ready to eat.
    b. Atleast 16 are ripe and ready to eat.
    c. Atmost 14 are ripe and **ready to eat**
5. If the probability is 0.2 that any one person will dislike the taste of new toothpaste, what is the probability that 5 of 18 randomly selected persons will dislike it?
6. If X is a Poisson random variable with mean 5, find (P(X=0), P(X=2), P(X≤11), P(X>25).

7. The number of customers arriving at a cafeteria is a Poisson RV with an average rate of 0.3 /minute. Find the probability that
   i) exactly 2 customers arrive in 10-minute span
   ii) 2 or more customers arrive in 10-minute span

8. The thickness of a flange on an aircraft component is uniformly distributed between 0.96 and 1.06 millimeters.
   (a) Determine the cumulative distribution function of flange thickness.
   (b) Determine the proportion of flanges that exceed 1.02 millimetres.
   (c) What thickness is exceeded by 90% of the flanges?
9. Suppose X has a uniform distribution over the interval [1.5, 5.5]. Determine (i) the mean and variance (ii) $P(X < 2.5)$ (iii) $P(X > 2 / X < 2.5)$ and (iv) $P(X > 1.5 / X < 2.5)$
10. The length of a telephone conversation in a booth has been exponentially distributed and found on average to be 5 minutes. Find the probability that a random call made from a booth (i) ends in less than 5 minutes. (ii) between 5 and 10 minutes.
11. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for 10 or more minutes.
12. Suppose X has an exponential distribution with a mean of 0.75, determine the following:
    (a) $P(X < 1.5)$ (b) $P(X \geq 1.5)$ (c) $P(-1<X<2)$
13. Suppose X has an exponential distribution with a mean equal to 10. Determine:
    (a) $P(X > 10)$ b) $P(X > 20)$ c) $P(X > 30)$ d) Find a such that $P(X < a) = 0.95$.
14. The amount of time that a watch will run without having to be reset is a random variable having an exponential distribution with a mean 120 days. Find the probability that such a watch will
    a. have to be set in less than 24 days and
    b. not have to be reset in at least 180 days.
15. The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with mean of 12.4 fluid ounces and a standard deviation of 0.1fluid ounce.
    a. What is the probability a fill volume is less than 12 fluid ounces?
    b. If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped?
16. The time it takes a cell to divide (called mitosis) is normally distributed with an average time of one hour and a standard deviation of 5 minutes.
    a. What is the probability that a cell divides in less than 45 minutes?
    b. What is the probability that it takes a cell more than 65 minutes to divide?
    c. What is the time it takes approximately 99% of all cells to complete mitosis?
17. The time to microwave a bag of popcorn using the automatic setting can be treated as a RV having a normal distribution with standard deviation 10 seconds. If the probability is 0.8212 that the bag will take less than 282.5 seconds to pop, find the probabilities that
    a. that it will take longer than 258.3 seconds to pop and
    b. that it will take on a value greater than 39.2 seconds.
18. Assume Z has a standard normal distribution. Use the standard normal table to find the value for z in each of the following cases.
19. (a) $P(-1.24 < Z < z) = 0.8$ (b) $P(-z < Z < z) = 0.95$ (c) $P(-z < Z < z) = 0.68$
20. Given that X has a normal distribution with $\mu = 200$ and $\sigma = 75$, find the probability that X assumes a value (a) greater than 300 (b) lesser than 170 (c) between 180 and 240
21. Assume X is a standard normal RV. Find the value of 'b' if $P(X<b) = 0.75$
22. Assume X is a standard normal RV. Find the value of 'c' if $P(X>c) = 0.002$
23. Given that X has a normal distribution with mean $\mu = 10$ and standard deviation $\sigma=3$.
    (a) If the probability that X assumes a value greater than b is 0.6, find b.
    (b) Also find $P(7<X<11)$.
24. Assume X is a normal RV with mean 15 and variance 4. Find the value of 'b' if $P(X<b) = 0.9$.
25. The mean and standard deviation of a certain group of 1000 high school grades that are normally distributed are 78% and 11% respectively. Find how many grades were above 90%?